WoSM Academy

MAT231-F27 · CALCULUS II

Functions of several variables

$\mathbb{R}^n$: norms, open and closed sets, limits and continuity

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WoSM Academy

Lecture 09 · Unit 9 · Fall 2027

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Where we are

- **Units 1–8**: series, power series, Taylor series, generating functions, Stirling's formula — all for functions of *one* variable.
- **Units 9–17**: differential calculus in $\mathbb{R}^n$. Partial derivatives, the gradient, the chain rule and the Jacobian, the Hessian, optimisation, gradient descent, Lagrange multipliers.
- Then multiple integrals (Units 18–22) and differential equations (Units 23–25).

Today we set the stage: distance in $\mathbb{R}^n$, which sets are open or closed, and what a limit means when you can approach a point from infinitely many directions. Vectors and inner products come from Linear Algebra (MAT102-S27).

This lecture

01 Functions of several variables

02 Norms and distance

03 Open and closed sets

04 Limits

05 Continuity

06 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 3* (OpenStax), Sections 4.1 and 4.2.

01

Functions of several variables

Examples

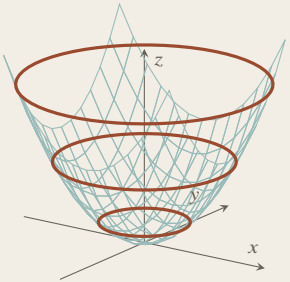
- $f(x, y) = x^2 + y^2$: $\mathbb{R}^2 \to \mathbb{R}$.
- The temperature $T(x, y, z)$ at a point of a room: $\mathbb{R}^3 \to \mathbb{R}$.
- The loss $L(\mathbf{w})$ of a machine-learning model with weights $\mathbf{w} \in \mathbb{R}^n$, often with n in the millions: $\mathbb{R}^n \to \mathbb{R}$.
- Polar coordinates $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$: $\mathbb{R}^2 \to \mathbb{R}^2$.

Examples

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- Polar coordinates $(r, \theta) \mapsto (r\cos\theta, r\sin\theta)$: $\mathbb{R}^2 \to \mathbb{R}^2$.

We write $\mathbf{x} = (x_1, \ldots, x_n)$ for a point of $\mathbb{R}^n$, and $f\colon D \to \mathbb{R}^m$ with a **domain** $D \subseteq \mathbb{R}^n$. For $m = 1$, f is **real-valued**; for $m > 1$, **vector-valued**, $f = (f_1, \ldots, f_m)$.

Graphs and level sets



- The **graph** of $f\colon \mathbb{R}^2 \to \mathbb{R}$ is the surface $z = f(x, y)$ in $\mathbb{R}^3$. Here $z = x^2 + y^2$, a paraboloid.
- A **level set** is $\{\mathbf{x} : f(\mathbf{x}) = c\}$. For $x^2 + y^2$ the level sets are circles of radius $\sqrt{c}$ (drawn at height c).
- Contour lines on a map are level sets of the height.

For $n \geq 3$ we cannot draw the graph; level sets and slices still help.

02

Norms and distance

The Euclidean norm

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ (Linear Algebra):

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i, \qquad \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{x_1^2 + \cdots + x_n^2}, \qquad \text{distance } \|\mathbf{x} - \mathbf{y}\|.$$

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- **Cauchy–Schwarz:** $|\mathbf{x} \cdot \mathbf{y}| \le \|\mathbf{x}\|\,\|\mathbf{y}\|$.
- **Triangle inequality:** $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$, since

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\,\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \le \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2.$$

- Hence also $\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \le \|\mathbf{x} - \mathbf{y}\|$.

Norms

Definition

A **norm** on $\mathbb{R}^n$ is a function $\mathbf{x} \mapsto \|\mathbf{x}\| \geq 0$ with (1) $\|\mathbf{x}\| = 0$ only for $\mathbf{x} = \mathbf{0}$, (2) $\|\lambda \mathbf{x}\| = |\lambda| \, \|\mathbf{x}\|$, (3) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

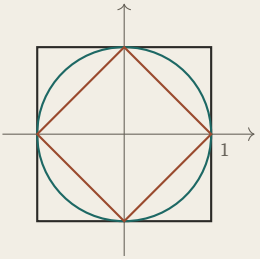
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- $\|\mathbf{x}\|_2 = \sqrt{\sum x_i^2}$: Euclidean (straight line).
- $\|\mathbf{x}\|_1 = \sum |x_i|$: Manhattan (along a street grid).
- $\|\mathbf{x}\|_\infty = \max_i |x_i|$: maximum (the moves of a chess king).

The “unit balls” $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$ in $\mathbb{R}^2$: a disc, a diamond, a square.



All norms on $\mathbb{R}^n$ are equivalent

Theorem

For every $\mathbf{x} \in \mathbb{R}^n$: $\quad \|\mathbf{x}\|_\infty \le \|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1 \le n\|\mathbf{x}\|_\infty$.

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Proof: $\max_i x_i^2 \le \sum x_i^2 \le (\sum |x_i|)^2$, and each $|x_i| \le \max_j |x_j|$.

- So if $\|\mathbf{x}_k - \mathbf{x}\| \to 0$ in one of these norms, it does in all of them. In fact any two norms on $\mathbb{R}^n$ satisfy such inequalities (we omit the proof).
- Everything in this lecture — open sets, limits, continuity — is the same for every norm. We use $\|\cdot\| = \|\cdot\|_2$ unless we say otherwise.
- Consequence: $\mathbf{x}_k \to \mathbf{x}$ if and only if each coordinate converges, $x_{k,i} \to x_i$ (use $\|\cdot\|_\infty$).

In infinite-dimensional spaces (functions, Lecture 4) norms need not be equivalent: uniform and pointwise convergence differ.

03

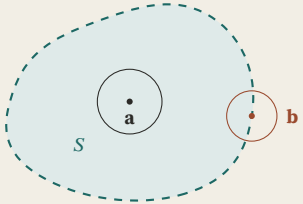
Open and closed sets

Balls, interior and boundary

- **Open ball:** $B(\mathbf{a}, r) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{a}\| < r\}$.
- $\mathbf{a}$ is an **interior point** of S if some ball $B(\mathbf{a}, r)$, $r > 0$, lies inside S.
- $\mathbf{a}$ is a **boundary point** of S if every ball $B(\mathbf{a}, r)$ contains points of S and points outside S.

Definition

S is **open** if all its points are interior points. S is **closed** if its complement $\mathbb{R}^n \setminus S$ is open.



a is interior; **b** is a boundary point.

Examples

- An open ball is open: if $\|\mathbf{x}-\mathbf{a}\| < r$, then $B(\mathbf{x},\varepsilon) \subseteq B(\mathbf{a},r)$ for $\varepsilon = r - \|\mathbf{x}-\mathbf{a}\|$, by the triangle inequality.
- The closed ball $\{\mathbf{x} : \|\mathbf{x}-\mathbf{a}\| \le r\}$ is closed. Its boundary is the sphere $\|\mathbf{x}-\mathbf{a}\| = r$.
- The half-plane $\{(x,y) : y > 0\}$ is open; $\{(x,y) : y \ge 0\}$ is closed.
- $[0,1)\times[0,1)$ is neither open nor closed.
- $\mathbb{R}^n$ and $\varnothing$ are both open and closed.

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> **Warning**
>
> "Not open" does not mean "closed". Most sets are neither.

Closed sets and sequences

> **Theorem**
>
> S is closed if and only if it contains the limit of every convergent sequence of points of S.

Proof idea: if $\mathbf{x}_k \in S$, $\mathbf{x}_k \to \mathbf{x}$ and $\mathbf{x} \notin S$, then $\mathbf{x}$ is in the open complement, so a ball around $\mathbf{x}$ misses S — but the $\mathbf{x}_k$ get into every ball around $\mathbf{x}$. The converse is similar.

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- S is **bounded** if it lies in some ball $B(\mathbf{0}, R)$.
- **Bolzano–Weierstrass in $\mathbb{R}^n$**: every bounded sequence has a convergent subsequence. (Apply the one-variable theorem of Calculus I, Unit 4, to one coordinate after another.)
- Closed and bounded sets are called **compact**. They are where continuous functions attain their maximum and minimum (Section 5).

04

Limits

Limits in $\mathbb{R}^n$

Definition

Let $f\colon D \to \mathbb{R}^m$ and let every ball around $\mathbf{a}$ contain points of D other than $\mathbf{a}$. Then $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$ if for every $\varepsilon > 0$ there is a $\delta > 0$ with

$$\|f(\mathbf{x}) - L\| < \varepsilon \quad \text{for all } \mathbf{x} \in D \text{ with } 0 < \|\mathbf{x} - \mathbf{a}\| < \delta.$$

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- The same words as in Calculus I, with $|\cdot|$ replaced by $\|\cdot\|$. The limit laws (sums, products, quotients) hold with the same proofs.
- A vector-valued f has a limit if and only if each component f_i does.
- New: $\mathbf{x}$ may approach $\mathbf{a}$ along infinitely many paths. The limit must be the same along all of them.

Different paths, different limits

Example ($f(x,y) = \dfrac{xy}{x^2+y^2}$ **at** $(0,0)$)

- Along the x-axis ($y = 0$): $f(x,0) = 0 \to 0$.
- Along the diagonal ($y = x$): $f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2} \to \frac{1}{2}$.
- Along $y = mx$: $f(x,mx) = \frac{m}{1+m^2}$: a different value for each slope.

So $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

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Strategy

To show that a limit does *not* exist, find two paths with different limits.

Lines are not enough

Example ($f(x,y) = \dfrac{x^2 y}{x^4 + y^2}$ **at** $(0,0)$)

- Along $x = 0$: $f = 0$. Along $y = mx$: $f(x, mx) = \frac{mx^3}{x^4 + m^2 x^2} = \frac{mx}{x^2 + m^2} \to 0$. Every line gives 0.
- Along the parabola $y = x^2$: $f(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2}$.

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The limit does not exist.

Warning

Checking all straight lines proves nothing. To show that a limit *exists*, you need an estimate that holds for all **x** near **a** at once.

Proving that a limit exists

Bound $|f(\mathbf{x}) - L|$ by something that depends only on $\|\mathbf{x} - \mathbf{a}\|$ and tends to 0.

Example ($\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^2+y^2} = 0$)

$x^2 \le x^2 + y^2$, so $\left|\frac{x^2 y}{x^2+y^2}\right| \le |y| \le \|(x,y)\| \to 0$.

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Example (Polar coordinates)

With $x = r\cos\theta$, $y = r\sin\theta$: $\frac{x^2 y}{x^2+y^2} = r\cos^2\theta\sin\theta$, and $|r\cos^2\theta\sin\theta| \le r \to 0$. But $\frac{xy}{x^2+y^2} = \cos\theta\sin\theta$ depends on θ: no limit.

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The polar bound must not depend on θ: for $\frac{x^2 y}{x^4+y^2}$ each fixed θ gives 0, yet there is no limit.

05

Continuity

Continuity

Definition

$f\colon D \to \mathbb{R}^m$ is **continuous at $\mathbf{a} \in D$** if $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$ (or $\mathbf{a}$ is an isolated point of D); it is continuous on D if it is continuous at every point.

- Equivalently: $\mathbf{x}_k \to \mathbf{a}$ implies $f(\mathbf{x}_k) \to f(\mathbf{a})$.
- The coordinate functions $\mathbf{x} \mapsto x_i$ are continuous. Sums, products, quotients (where the denominator is not 0) and compositions of continuous functions are continuous.
- So polynomials and rational functions in $x_1, \ldots, x_n$ are continuous where defined, and so are e^{xy}, $\sin(x^2 + y)$, and $\mathbf{x} \mapsto \|\mathbf{x}\|$ (as $\big|\|\mathbf{x}\| - \|\mathbf{a}\|\big| \le \|\mathbf{x} - \mathbf{a}\|$).

Continuous in each variable is not enough

Example

$f(x,y) = \frac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$.

- For each fixed y, $x \mapsto f(x,y)$ is continuous; for each fixed x, $y \mapsto f(x,y)$ is continuous. (At 0: $f(x,0) = 0$ and $f(0,y) = 0$.)
- But f is not continuous at $(0,0)$: along $y = x$ it equals $\frac{1}{2}$.

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- But f is not continuous at $(0,0)$: along $y = x$ it equals $\frac{1}{2}$.

Continuity in $\mathbb{R}^n$ is about all ways of approaching a point, not only the coordinate directions. We will meet the same lesson for derivatives next lecture.

Continuity and open and closed sets

> **Theorem**
>
> If $f\colon \mathbb{R}^n \to \mathbb{R}$ is continuous and $c \in \mathbb{R}$, then $\{\mathbf{x} : f(\mathbf{x}) < c\}$ is open, and $\{\mathbf{x} : f(\mathbf{x}) \le c\}$ and $\{\mathbf{x} : f(\mathbf{x}) = c\}$ are closed.

Proof: if $f(\mathbf{a}) < c$, continuity gives a ball around $\mathbf{a}$ on which $f < c$. The other two sets have open complements $\{f > c\}$ and $\{f < c\} \cup \{f > c\}$.

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Example

- $\{(x, y) : x^2 + y^2 < 1\}$ is open; the unit circle $x^2 + y^2 = 1$ is closed.
- $\{\mathbf{x} : x_1 \ge 0, \ldots, x_n \ge 0\}$ is closed: an intersection of closed sets.
- The level set $\{(x, y) : x^2 - y^2 = 1\}$, a hyperbola, is closed.

The extreme value theorem

Theorem

A continuous function $f\colon K \to \mathbb{R}$ on a nonempty closed and bounded set $K \subseteq \mathbb{R}^n$ attains a maximum and a minimum on K.

Proof idea, as in Calculus I: take $\mathbf{x}_k \in K$ with $f(\mathbf{x}_k) \to \sup_K f$; a subsequence converges (Bolzano–Weierstrass) to a point of K (K is closed), and f is continuous there.

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- Both conditions matter: $f(x, y) = x$ has no maximum on the open disc, nor on the closed but unbounded half-plane $x \geq 0$.
- Unit 14 finds such extrema with derivatives; Unit 16 does it on curved sets such as circles, with Lagrange multipliers.

06

Summary and practice

Summary

- Functions $f\colon D \subseteq \mathbb{R}^n \to \mathbb{R}^m$; graphs and level sets.
- Norms $\|\cdot\|_2$, $\|\cdot\|_1$, $\|\cdot\|_\infty$; on $\mathbb{R}^n$ all norms are equivalent, so convergence means convergence of each coordinate.
- Open sets (every point has a ball inside), closed sets (complement open, or: contain their limits), compact sets (closed and bounded).
- Limits must agree along *every* path; two paths with different limits disprove a limit; a bound in terms of $\|\mathbf{x} - \mathbf{a}\|$ proves one.
- Continuous functions: built from coordinates by algebra and composition; $\{f < c\}$ is open, $\{f \le c\}$ closed; extreme values on compact sets.

Practice

1. Describe the level sets of $f(x, y) = x^2 - y^2$ and of $g(x, y) = xy$.
2. Show that $\|\mathbf{x}\|_1 \le \sqrt{n}\,\|\mathbf{x}\|_2$ for $\mathbf{x} \in \mathbb{R}^n$.
3. Open, closed, both or neither? (a) $x^2 + y^2 \le 1$ (b) $0 < x < 1$ in $\mathbb{R}^2$ (c) $y = x^2$ (d) $x > 0,\ y \ge 0$ (e) $\mathbb{Q}^2$ (f) $\varnothing$
4. Find the limit at $(0, 0)$, or show that it does not exist: (a) $\frac{x^2 - y^2}{x^2 + y^2}$ (b) $\frac{xy^2}{x^2 + y^2}$ (c) $\frac{\sin(x^2 + y^2)}{x^2 + y^2}$ (d) $\frac{xy^2}{x^2 + y^4}$
5. Where is $f(x, y) = \frac{x^2 + y^2}{x - y}$ continuous?
6. Show that $\mathbf{x} \mapsto \|\mathbf{x}\|$ is continuous on $\mathbb{R}^n$.
7. Give a continuous function on a bounded set in $\mathbb{R}^2$ that has no maximum.

Answers

1. Hyperbolas $x^2 - y^2 = c$ (and the lines $y = \pm x$ for $c = 0$); hyperbolas $xy = c$ (and the two axes for $c = 0$).
2. Cauchy–Schwarz with $(|x_1|, \dots, |x_n|)$ and $(1, \dots, 1)$.
3. (a) closed (b) open (c) closed (a level set) (d) neither (e) neither: no ball fits inside, and irrational points are limits of rational ones (f) both.
4. (a) does not exist: 1 along the x-axis, -1 along the y-axis. (b) 0: $|\cdot| \le |x|$. (c) 1: $\frac{\sin t}{t} \to 1$ with $t = x^2 + y^2 \to 0$. (d) does not exist: 0 along lines, $\frac{1}{2}$ along $x = y^2$.
5. Wherever $x \neq y$: a quotient of continuous functions.
6. $\big|\|\mathbf{x}\| - \|\mathbf{a}\|\big| \le \|\mathbf{x} - \mathbf{a}\|$, so take $\delta = \varepsilon$.
7. $f(x, y) = x$ on the open unit disc: its values approach 1 but never reach it.

Next time

Lecture 10 — Partial derivatives and differentiability (Unit 10)

- Partial derivatives: differentiate in one variable, freeze the others
- Why partial derivatives are not enough
- The total derivative: the best linear approximation, and the Jacobian matrix

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 4.3 and 4.4.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.